

Carbon and its Compounds

Science • Chapter 4 • Chapter Summary & Study Guide for Daksh

Carbon is everywhere — in your food, your clothes, your fuel, your DNA. Yet it forms only ~0.02% of Earth's crust. The reason carbon dominates the chemistry of life is its unique ability to form a vast number of stable compounds. This chapter explains why carbon is special, how its compounds are named (IUPAC), what reactions they undergo, and looks at two important compounds in detail — **ethanol** and **ethanoic acid**.

Why carbon forms so many compounds

- **Catenation** — carbon atoms link to other carbon atoms forming long chains, branched chains and rings. Silicon shows some catenation but with less stability.
- **Tetravalency** — each carbon has 4 valence electrons and forms 4 covalent bonds. This lets carbon bond with H, O, N, S, halogens, and itself in many shapes.
- **Strong covalent C–C bonds** make these compounds stable; **shared pairs** with H, O, N create molecules of life.

Bonds, types and homologous series

Carbon almost always forms **covalent bonds** by sharing electrons (single, double, triple). Saturated carbon compounds have only single C–C bonds (**alkanes**); unsaturated ones have double (**alkenes**) or triple (**alkynes**) bonds. Unsaturated compounds are more reactive and undergo **addition** reactions; saturated ones undergo **substitution** reactions.

Hydrocarbon families (general formulae):

Alkanes	$C_nH_{(2n+2)}$	– single bonds	e.g. CH_4 (methane), C_2H_6 (ethane)
Alkenes	$C_nH_{(2n)}$	– one C=C	e.g. C_2H_4 (ethene), C_3H_6 (propene)
Alkynes	$C_nH_{(2n-2)}$	– one C≡C	e.g. C_2H_2 (ethyne / acetylene)

Functional groups:

-OH alcohol	-CHO aldehyde	-CO- ketone	-COOH carboxylic acid
-X (halo)	-NH ₂ amine	-O- ether	

A **homologous series** is a family of carbon compounds where successive members differ by $-CH_2-$ and have similar chemical properties (e.g. methanol, ethanol, propanol, ...). **Functional groups** determine the characteristic chemistry of a homologous series.

IUPAC nomenclature uses a root (counts the carbons), a suffix for the family, and prefixes for branches/substituents. e.g. CH_3-CH_2-OH = ethan + ol = **ethanol**; CH_3-COOH = ethan + oic acid = **ethanoic acid**.

Important reactions of carbon compounds

- **Combustion:** Carbon compounds burn in oxygen to give $CO_2 + H_2O + \text{heat} + \text{light}$. Saturated → clean blue flame; unsaturated → yellow sooty flame (incomplete combustion).
- **Oxidation:** Alcohol → Carboxylic acid using oxidising agents (alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$). e.g. $CH_3CH_2OH + [O] \rightarrow CH_3COOH$.
- **Addition reactions:** Unsaturated compounds + $H_2 \rightarrow$ saturated (in presence of Ni catalyst). Used in *hydrogenation of vegetable oils* to make vanaspati.
- **Substitution:** Alkanes + Cl_2 (in sunlight) → chloroalkanes + HCl. e.g. $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$.

Ethanol and Ethanoic acid

Ethanol (CH_3CH_2OH) is the alcohol in alcoholic drinks. It is a colourless liquid, soluble in water, neutral to litmus. With sodium → sodium ethoxide + H_2 . With hot conc. H_2SO_4 at 443 K → ethene + H_2O (dehydration). Excessive intake harms health and is widely abused.

Ethanoic acid (CH_3COOH), commonly known as **vinegar** in dilute form, has a characteristic smell and freezes a little above room temperature (called *glacial acetic acid*). It is a weak acid. Reactions: with NaOH →

$\text{CH}_3\text{COONa} + \text{H}_2\text{O}$; with $\text{NaHCO}_3 \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O} + \text{CO}_2 \uparrow$; **esterification** with ethanol (and an acid catalyst) $\rightarrow \text{CH}_3\text{COOC}_2\text{H}_5$ (ethyl ethanoate, a sweet-smelling ester used in flavours and perfumes).

Soaps, detergents, micelles

Soap is the sodium/potassium salt of long-chain fatty acids. A soap molecule has a hydrophilic ionic head ($-\text{COO}^-\text{Na}^+$) and a long hydrophobic hydrocarbon tail. In water it forms **micelles** — clusters with tails turned inward into the oily dirt and heads outward in water. Rinsing carries the dirt-loaded micelles away. **Hard water** contains $\text{Ca}^{2+}/\text{Mg}^{2+}$ ions that form scum with soap; **detergents** (synthetic salts of long-chain sulphonic acids) work even in hard water because their calcium/magnesium salts are soluble.

★ MUST-REMEMBER POINTS

- Catenation + tetravalency = vast organic chemistry.
- Saturated $\rightarrow$ substitution; Unsaturated $\rightarrow$ addition.
- Homologous series differ by $-\text{CH}_2-$ and ~ 14 g/mol; share a functional group.
- Vanaspati comes from hydrogenation of vegetable oils (Ni catalyst).
- Combustion: blue flame (saturated, complete); sooty flame (unsaturated, incomplete).
- Esterification: alcohol + carboxylic acid $\rightarrow$ ester + water (sweet-smelling).
- Soap micelles trap oily dirt; detergents work even in hard water.

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